Engineering Physics.

Q.1 superconductors :-

Superconductors are the materials that exhibit zero electrical resistance and the expulsion of magnetic fields (the meissner effect) when cooled below a specific characteristics temp known as the critical temp (Tc)

Properties :

1. Zero electrical Resistance
2. Meissner effect
3. Critical temperature

Q.2. For Fcc crystal :

Lattice constant and atomic radius :

$a = 2\sqrt{2}\,R$

$R = 1.441\ \text{Å}$

$a = 2\sqrt{2} \times 1.44$

$a = 4.075$

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

for (220)

$$d_{220} = \frac{4.075}{\sqrt{2^2 + 2^2 + 0^2}}$$

$$\boxed{d_{220} = 1.44}$$

3.

Feature	Type I	Type II
Magnetic field	Completely expel magnetic field	Partially expel magnetic field
Critical field	One critical field	Two critical fields
Transition	sudden transition	Gradual transition
Materials	Pure metals	Alloys and compounds
Example	Mercury, Tin	Niobium - Titanium

4. In BCC, 8 corners of the cube.
1 atom at the center of the cube.

Total atoms:

$= 8 \times \frac{1}{8} + 1$

$= 1 + 1$

$= 2$

Effective no. of atoms in BCC = 2 atoms per unit cell.